

Two kinds of conditionals, and two analyses

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1. Indicative versus counterfactual conditionals

A famous contrast (Adams 1970):

- (1) If Oswald didn't kill Kennedy, someone else did
- (2) If Oswald hadn't killed Kennedy, someone else would have

Insofar as we are buying into the two-proposition model, it's clear that these involve the same two propositions: the proposition that Oswald didn't kill Kennedy and the proposition that someone other than Oswald killed Kennedy. But they mean very different things.

(1) is called an *indicative* conditional; (2) a *counterfactual* or *subjunctive* conditional. We'll use the labels 'counterfactual' and 'subjunctive' interchangeably.

- A problematic suggestion: what's semantically distinctive about (2) is that it entails, or suggests, that it is *false* that Oswald killed Kennedy. Evidence against this (Anderson 1951):
- (3) If Jones had taken arsenic, he would have shown just exactly those symptoms which he does in fact show.

Thus 'counterfactual' is something of a misnomer.

- But so is 'subjunctive': (2) does not use the subjunctive mood (which is pretty vestigial in English). Its grammatical hallmarks are, rather: (a) the presence of 'would' in the main clause, and (b) an additional level of past tense in the 'if' clause. Sabrina Iatridou (2000) calls (b) "fake past tense", and finds analogues in many other languages — see section 6.2 of the von Stechow handbook article.
- There is a debate about whether to classify (4) along with (1) or along with (2):
- (4) If Oswald doesn't kill Kennedy, someone else will.

See Bennett §6. Given this controversy, let's try whenever possible to avoid relying on such 'does-will' conditionals as examples.

Incidentally: the following contrast may help shed light on this dispute:

- (5) a. If you were given a million dollars tomorrow, you would be richer than you will ever actually be.
- b. ? If you are given a million dollars tomorrow, you will be richer than you will ever actually be.

- It's worth considering what happens when we use different combinations of tenses and 'would' marking:

- (6)
- If Oswald hadn't killed Kennedy, someone else had
 - If Oswald hadn't killed Kennedy, someone else would
 - If Oswald didn't kill Kennedy, someone else would
 - If Oswald wasn't killing Kennedy, someone else was

Are these more like (1) or more like (2)?

2. Two special uses of conditionals

von Fintel characterises indicatives and subjunctives as two of four categories, the others being 'biscuit' uses as in (7), and 'premise' uses as in (8):

- (7)
- There are biscuits on the sideboard if you want some (Austin)
 - If you don't mind my saying so, this is boring.

- (8) If it is indeed that late, we should leave

(I'm not sure (8) is really a good example of what he's trying to get at here.) Unlike von Fintel, we will use the label 'indicative' in such a way as to encompass sentences like (7) and (8). It is not obvious that there is anything special about them from the point of view of semantics (as opposed to pragmatics). But until we have looked into the matter further we should be open to the idea that there is.

3. The material conditional analysis

A proposal: the proposition expressed by 'If ϕ , ψ ' (in a context) is identical to the one expressed by 'Either not- ϕ , or ψ ' (in the same context). Or at least, these propositions are "equivalent" in the sense that each entails the other.

- We stipulatively define ' $\supset$ ' as the material conditional, i.e. ' $\phi \supset \psi$ ' means the same as 'either not- ϕ or ψ '.

This theory could be applied to indicatives, counterfactuals, or both. Also, we could hold that within one of the two classes, some conditionals are material and others are not.

For counterfactuals, the material conditional analysis looks *insane*, and as far as I am aware it has no current defenders. It's easy to come up with examples of counterfactuals with false antecedents that seem obviously false:

- (9) If I had snapped my fingers I would have disappeared

By contrast, the claim that indicative conditionals are material conditionals has some distinguished defenders (although it is very much a minority view).

Why the contrast?

- Perhaps it isn't so easy to come up with counterexamples as glaring as (9)? (What do we think?)
- In the case of indicatives, the **or-to-if inference**—from ' ϕ or ψ ' to 'if not ϕ , then ψ '—seems *prima facie* valid, whereas the inference from ' ϕ or ψ ' to 'if it had been that not- ϕ , it would have been that ψ ' looks atrocious in many cases. But if the or-to-if inference really is valid for indicative conditionals, it is very hard to avoid the conclusion that they are material conditionals.

On the other hand, the view that indicatives are material conditionals faces many serious objections.

- Famous objections concern arguments that come out valid according to the view, but that look kind of weird: the so-called 'paradoxes of material implication'. First 'paradox': the argument from 'not ϕ ' to 'if ϕ then ψ '. Second 'paradox': the argument from ψ to 'if ϕ then ψ '.
- One that I think is especially serious: often it seems appropriate to be *much less confident* that if ϕ , ψ than that $\phi \supset \psi$. For example: you should be very confident that either you will never win the lottery or you will be elected President (since you could be very confident that you will never win the lottery). But you shouldn't be at all confident that if you win the lottery you will be elected President. (Right?)

We will return later to the debate about whether indicative conditionals are material conditionals.

Reading for next week:

- Lewis, *Counterfactuals*, chapter 1 (on NYU Classes)
- Bennett, chapter 10